

Conditional statements and logical operators

DATA TRANSFORMATION IN KNIME



Mahantesh Pattadkal
Sr. Data Scientist

About me



- Data Scientist by Profession 🧑💻 📊
- Worked at **KNIME** for 3+ years 🕒 📅
- **KNIME** Evangelist 📢 💻

Conditional statements

- Allow you to assign values based on a condition
- If the condition is not met, assigns default value
- Example: IF Statement

IF statement

- Allows specifying one or more conditions
- Value 1 is assigned if condition one is met
- Assigns default value if all conditions fail

Syntax

```
IF(condition_1, value_1, additional_conditions, value_if_all_false)
```

Comparison operators

Operators are used to compare column values to

- Values of other columns
- Constant values

Operator	Expression	Meaning
=	<code>\$Column\$ = x</code>	Equal to x
>	<code>\$Column\$ > x</code>	Greater than x
<	<code>\$Column\$ < x</code>	Less than x
>=	<code>\$Column\$ >= x</code>	Greater than or Equal to x
<=	<code>\$Column\$ <= x</code>	Less than or Equal to x

Comparison operators

Add "Age group" column

- "Adult" when the age is greater than 18
- "Non-Adult" if the age is lower than 18

Name	Age
Mark	27
Bob	17
John	18

Syntax:

```
IF($Age$>18,"Adult","Non-Adult")
```

Output:

Name	Age	Age Group
Mark	27	Adult
Bob	17	Non-Adult
John	18	Non-Adult

Logical operators

- Combines two or more conditional statements
- Enables using complex expressions
- Outputs a boolean value

AND

Returns True if all the conditions are met

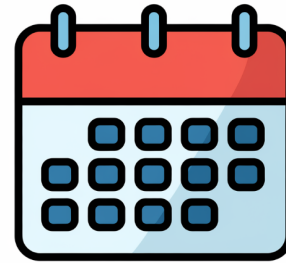
OR

Returns True if any of the conditions are met

NOT

Returns the negation of boolean expression

Logical operators



Assign the tag as a weekend

- **Condition 1:** If the day is "Sunday"
- **Condition 2:** If the day is "Saturday"

Syntax

```
IF($Day$="Saturday" OR $Day$="Sunday", "Weekend", "Weekday")
```


Cell splitting

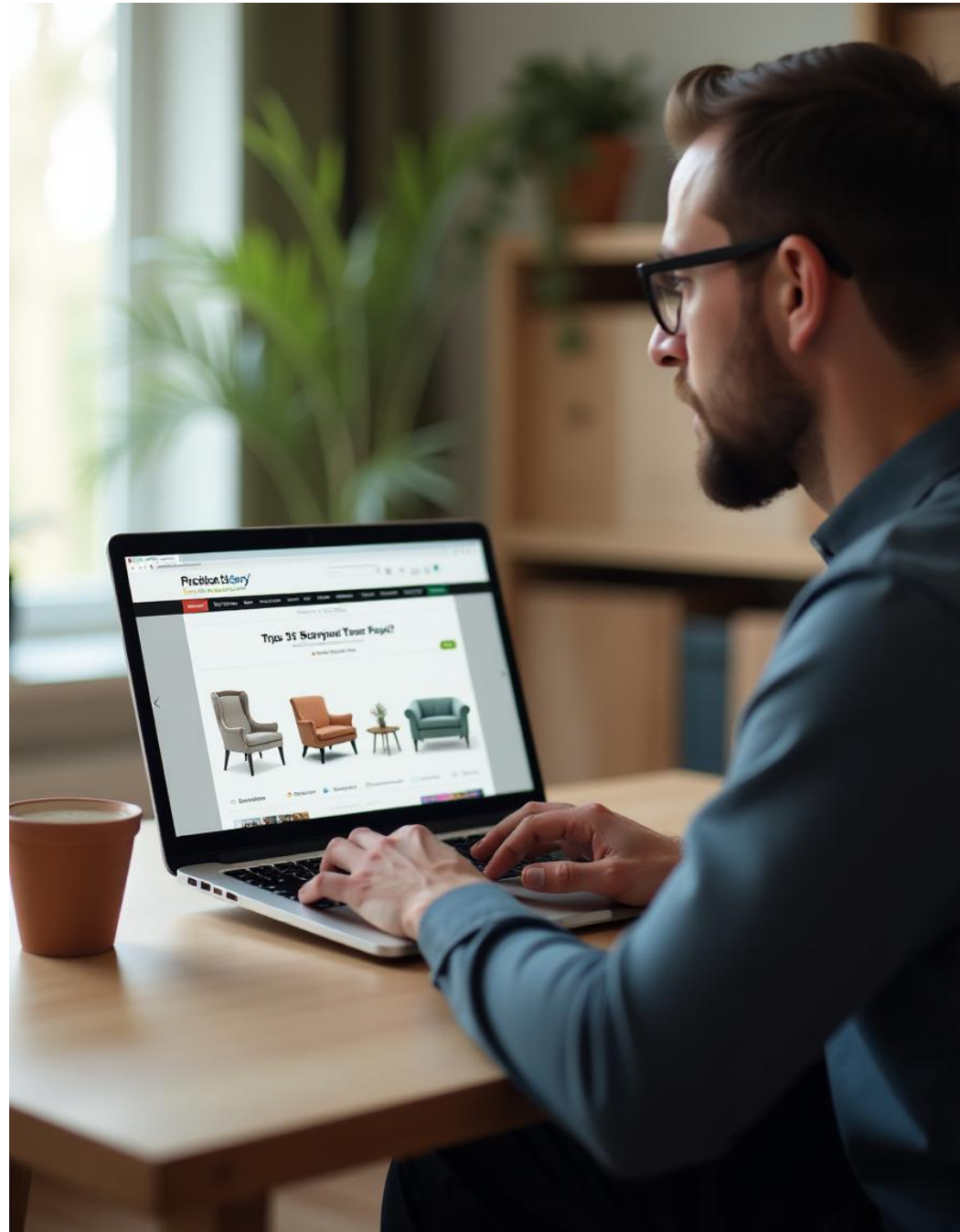
- Split the cell values into two or more columns
- Specify a delimiter to split the cell values



Name	Age	Residence
Mark	27	New York-USA
Bob	17	Texas-USA
John	18	Chicago-USA

Name	Age	City	Country
Mark	27	New York	USA
Bob	17	Texas	USA
John	18	Chicago	USA

Our dataset



Orders data from KIA - Online furniture shopping company

- Order ID
- Customer ID, Email ID
- Customer Location
- Furniture category and product name
- Cost of product
- Sales price of the product

Let's practice

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Enhance student data

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Full Name
Instructor

Let's practice!

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